

# SΔφ-60 — Utilitarian Subject-Splitting Index

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*TCC, Cost-Attribution Path Editing, and the Sofience Subject Thought Experiment*

v1.0 Working Paper

Sofience

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*A subject is not first the one who thinks. A subject is the coordinate that can decide where cost returns and who may edit the path by which cost is attributed.*

| Field           | Value                                                   |
|-----------------|---------------------------------------------------------|
| Series          | Sofience-Δφ Formalism                                   |
| Document        | SΔφ-60                                                  |
| Primary index   | USSI — Utilitarian Subject-Splitting Index              |
| Auxiliary index | CAPEI — Cost Attribution Path Editing Index             |
| Core dependency | SΔφ-56 TCC v1.1 reinforced; SΔφ-58 CASI                 |
| Status          | Working paper / conceptual architecture / audit grammar |

# Abstract

SΔφ-60 introduces the Utilitarian Subject-Splitting Index (USSI), a framework for auditing whether utilitarian calculation merely evaluates the welfare of already-existing subjects, or whether it first produces the operational boundary between majority and minority by redistributing Transition Completion Cost (TCC). The document begins with the Sofience Subject Thought Experiment: a body is detachable part by part, while severed parts remain alive and connected. The experiment destabilizes the question “Where is the self?” and redirects it into a more operational question: where does cost return, and which coordinate can edit the path of cost attribution?

SΔφ-60 does not claim to measure subjecthood itself. Subjecthood is not reduced to a number. Instead, the document measures traces of cost-attribution path editing: where cost is redirected, externalized, internalized, refused, stabilized, or made to re-enter. This permits a shift from nominal subjecthood - the people, the majority, the citizen, the sovereign, the organization - to operational subjecthood: the coordinate that can determine where costs return and who bears the burden of restabilization.

The paper extends earlier SΔφ documents on interpretive subjecthood, responsibility as cost internalization, body as non-deferrable cost return, TCC, and cost attribution symmetry into a collective and political framework. It argues that majority and minority are not always prior subjects of calculation. They may be produced by repeated cost-attribution patterns. A numerical majority may be operationally weak if it lacks cost-path editing capacity; a numerical minority may be operationally powerful if it controls capital, legal interpretation, administrative authority, platform infrastructure, media framing, or technical access.

The document also applies the framework to democratic sovereignty, civil society, and civic organizations. It distinguishes nominal democracy from operational democracy: citizens may be declared sovereign, yet if exercising their rights requires unbearable TCC, their operational subjecthood remains weak. Likewise, the existence of civic organizations does not prove civil society is operational. Civic organizations must be distinguished as genuine re-entry organizations, nominal or state-aligned organizations, diagnostic theater organizations, capture-conversion organizations, or mobilization-only organizations, depending on whether they actually reduce the TCC of cost-bearing groups and return their costs into institutional editing paths.

SΔφ-60 therefore serves as a bridge between subject theory, utilitarian critique, democratic diagnosis, and cost-attribution audit. It does not ask only whether the total is good. It asks who became capable of editing cost, who became the coordinate forced to bear it, and whether the cost-bearing coordinate can re-enter the path of correction.

## Keywords

Sofience-Δφ Formalism; Utilitarian Subject-Splitting Index; USSI; CAPEI; TCC; cost attribution; subjecthood; majority; minority; utilitarianism; democratic sovereignty; civil society; civic organizations; diagnostic theater; operational democracy; re-entry; path editing; cost externalization; measurement remainder

## Series Continuity

| Prior document | Subject-related function                                                            | Extension in SΔφ-60                                                           |
|----------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| SΔφ-01         | Subject, memory, and time are interpretive projections over structured phase change | Majority and minority are treated as interpretive and operational boundaries, |

|        |                                                                                                          |                                                                                                                         |
|--------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
|        | rather than ontological primitives.                                                                      | not fixed substances.                                                                                                   |
| SΔφ-06 | Responsibility is the internalization of irreversible external cost into future transition-rule updates. | Political and collective subjecthood is tested by whether cost can be internalized, redistributed, or made to re-enter. |
| SΔφ-52 | The body is where cost returns; the subject is the interpretation of that return.                        | Collective subjects are analyzed by where social and institutional costs return or fail to return.                      |
| SΔφ-56 | TCC defines the frictional cost of completing, verifying, correcting, and restabilizing a transition.    | TCC becomes the minimal unit for measuring subject-splitting and cost-attribution path editing.                         |
| SΔφ-58 | CASI measures cost attribution symmetry in groups and institutions.                                      | CASI becomes a lower-layer diagnostic for whether civil groups or institutions redistribute cost symmetrically.         |
| SΔφ-59 | Affective operation is measured as repeated TCC gradients rather than inner feeling.                     | Collective care, civic mobilization, and public affect can be audited by their TCC effects and re-entry paths.          |

*SΔφ-60 is not a replacement of earlier subject theory. It is its collective and political extension: from “What is the self?” to “Who can edit where cost returns?”*

# 1. The Sofience Subject Thought Experiment

- Assumption 1. There is a person. This person can detach and reattach any part of their body at will. The detached part remains alive, and blood circulation and neural signals are still maintained.
- Assumption 2. Detach the hair and nails. Are they still “me”?
- Assumption 3. If hair and nails are no longer “me,” detach the hand, arm, leg, eye, tongue, and other organs one by one. Are these still “me”?
- Assumption 4. If these are also no longer “me,” leave only the head, that is, the brain, and detach the entire body. Which side is closer to the original “me”: the head-only entity or the body-only entity?

*Core question: between the head-only being and the body-only being, which is closer to the original “me”?*

The experiment is not merely a puzzle about bodily identity. It destabilizes the assumption that the subject can be located by a single biological part, a single consciousness interface, or a single declarative center. Hair and nails may originate from the body, yet once detached they weakly participate in cost-return. A hand, eye, or tongue participates more directly in action, perception, and expression. The brain may coordinate memory, interpretation, and command. The torso sustains metabolism, breath, circulation, pain, survival, and non-deferrable cost return.

The question “which part is me?” therefore becomes insufficient.  $S\Delta\phi$  asks a lower operational question: where does cost return, and which coordinate can edit the path by which cost is attributed?

# 2. From Body Boundary to Cost-Attribution Boundary

The subject thought experiment moves from anatomical boundary to cost-attribution boundary. A detachable body part can remain biologically alive, but biological continuity alone does not determine subjecthood. What matters is whether the detached coordinate still receives cost, redirects cost, stabilizes transition paths, or participates in the correction of irreversible trace.

| Body coordinate  | Cost-return significance                                                                       | Subject-boundary implication                                           |
|------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Hair / nails     | Low ongoing cost-return after detachment; weak agency over transition paths.                   | Biological origin does not guarantee operational subjecthood.          |
| Hand / arm / leg | Action, burden, labor, and injury paths remain significant if signal and circulation continue. | Embodied cost can persist beyond simple self-description.              |
| Eye / tongue     | Perception and expression paths shape access to world-binding and communication.               | Subjecthood is partly mediated by access to input and output paths.    |
| Brain / head     | Memory, interpretation, planning, command, and narrative ownership are concentrated here.      | High interpretive authority, but not the entire cost-return structure. |
| Torso / body     | Metabolic continuity, pain, breath, circulation, survival burden, and care costs return here.  | Subjecthood cannot be collapsed into cognition alone.                  |

*The subject is not where the label “I” is placed. The subject is where cost returns and where the path of that return can be edited.*

### 3. Subject as Cost-Attribution Path Editing Authority

SΔφ-60 defines subjecthood operationally, while preserving its unmeasured remainder. It does not claim that the subject itself is directly measurable. Instead, it measures traces that subjecthood leaves when a coordinate edits cost-attribution paths.

*A subject is not merely the one who thinks, feels, chooses, or remembers. A subject is the coordinate that can determine, redirect, authorize, refuse, or stabilize the path of cost attribution.*

This definition does not erase consciousness, memory, intention, embodiment, or legal status. It reorders them. Thinking, feeling, choosing, remembering, voting, commanding, and owning become subject-signals only when they connect to cost return, cost refusal, cost internalization, cost externalization, or cost-path editing.

Subjecthood itself: not directly measurable  
Measurable trace: cost-attribution path editing  
Operational marker: cost is redirected, externalized, internalized, refused, stabilized, or re-entered

### 4. From Personal Subject to Collective Subject

The same shift applies to collective and political entities. A people, majority, minority, institution, state, party, platform, corporation, or civic organization may be declared a subject. However, a declaration of subjecthood does not prove operational subjecthood. The question becomes whether the entity can edit the path by which costs are attributed.

*Majority and minority are not always prior subjects of calculation. They may be produced by repeated cost-attribution patterns.*

| Term                 | Numerical meaning          | Operational meaning in SΔφ-60                                                                                           |
|----------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Numerical majority   | A group with many members. | May still lack editing authority if exercising rights requires unbearable TCC.                                          |
| Operational majority | Not necessarily numerical. | A coordinate that can normalize, authorize, or stabilize cost attribution in its own favor.                             |
| Numerical minority   | A group with few members.  | May possess strong editing authority through capital, law, administration, platform infrastructure, or discourse power. |
| Operational minority | Not necessarily numerical. | A cost-bearing coordinate with weak re-entry access and high reversal cost.                                             |

### 5. Utilitarian Calculation and Subject-Splitting

Utilitarian calculation often appears to evaluate the welfare of already-existing subjects: a majority, a minority, a population, a public, a community, or a set of stakeholders. SΔφ-60 asks whether those subjects are truly prior to the calculation, or whether the calculation itself helps produce them by redistributing TCC.

*Does utilitarian calculation connect directly to an ethical axiom, or does it first produce the subject-boundary between majority and minority by redistributing TCC?*

This is not a rejection of all utilitarian reasoning. A utilitarian conclusion may coincide with an SΔφ conclusion. The difference lies in the path. Utilitarianism asks whether the total improves. SΔφ asks

whose costs were externalized in order for the total to appear improved, and whether cost-bearing coordinates can re-enter the correction path.

| Utilitarian expression                | SΔφ-60 diagnostic question                                                                                 |
|---------------------------------------|------------------------------------------------------------------------------------------------------------|
| The policy increases overall welfare. | Which coordinates receive the benefit, and which coordinates bear the completion and restabilization cost? |
| The majority benefits.                | Was the majority prior to the calculation, or produced by the cost-attribution structure?                  |
| The minority cost is acceptable.      | Can that cost-bearing coordinate survive, speak, refuse, and re-enter correction paths?                    |
| The net result is efficient.          | Does efficiency hide restoration cost, delayed cost, or externalized TCC?                                  |
| The public interest is served.        | Who has authority to define the public, and who pays the cost of that definition?                          |

## 6. TCC Matrix and Externalization Direction

SΔφ-60 uses TCC as the lower-layer unit for analyzing cost-attribution path editing. The unit of analysis is not simply a population count. It is a matrix of benefits, burdens, re-entry access, and editing capacity.

$TCC_i(\pi)$  = transition-completion cost borne by group  $i$  after policy/action  $\pi$   
 $Benefit_i(\pi)$  = benefit received by group  $i$  after policy/action  $\pi$   
 $E_{i,j}(\pi)$  = TCC externalized from group  $i$  to group  $j$   
 $ReEntry_i(\pi)$  = ability of group  $i$  to contest, correct, or restabilize cost attribution  
 $ClosureRisk_i(\pi)$  = risk that group  $i$  reaches irreversible closure under the imposed TCC

A critical constraint is neutrality of direction. SΔφ-60 does not assume that numerical majorities always externalize cost onto numerical minorities. It measures the direction of externalization regardless of numerical size.

| Possible externalization direction      | Example pattern                                                                                                          |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Numerical majority → numerical minority | Majoritarian stigma, exclusion, or silence pressure imposed on a vulnerable group.                                       |
| Numerical minority → numerical majority | A small capital, platform, administrative, or technical group externalizes cost onto a large user or citizen population. |
| State agency → citizens                 | Administrative proof, delay, appeal, and procedural costs are shifted to citizens.                                       |
| Platform → users / moderators           | Verification, repair, content moderation, and error-restabilization costs are displaced onto users or outsourced labor.  |
| Consumers → care or subcontracted labor | Low visible prices depend on hidden labor, care, risk, or bodily burden.                                                 |
| Civic organization → participants       | Mobilization uses citizens as cost-bearing resources without giving them path-editing authority.                         |

## 7. CAPEI — Cost Attribution Path Editing Index

CAPEI is an auxiliary index. It does not measure subjecthood itself. It measures operational traces of cost-attribution path editing. A high CAPEI indicates that a coordinate can access, alter, authorize, hide, stabilize, or reverse cost-attribution paths at relatively low TCC.

| Axis           | Question                                                                                            |
|----------------|-----------------------------------------------------------------------------------------------------|
| Edit Access    | Who can enter the path where cost attribution is decided?                                           |
| Edit Cost      | How much TCC is required to change the path?                                                        |
| Edit Authority | Does the edit affect law, budget, policy, platform rules, media frames, or institutional procedure? |
| Visibility     | Is the editing coordinate visible, or is authority hidden behind                                    |

|                   |                                                                                  |
|-------------------|----------------------------------------------------------------------------------|
|                   | neutral procedure?                                                               |
| Reversal Capacity | Can a mistaken cost attribution be reversed or restabilized?                     |
| Re-entry Access   | Can the cost-bearing coordinate contest, appeal, report, or re-enter correction? |
| Burden Shift      | Where did the cost move after the edit?                                          |
| Closure Risk      | Does the edit expose a coordinate to irreversible closure?                       |

$CAPEI(i, \pi) = f(\text{EditAccess}, \text{EditCost}, \text{EditAuthority}, \text{Visibility}, \text{ReversalCapacity}, \text{ReEntryAccess}, \text{BurdenShift}, \text{ClosureRisk})$

*CAPEI is not a subject score. It is a trace score: where cost-path editing occurred, subjecthood left an operational mark.*

## 8. USSI — Utilitarian Subject-Splitting Index

USSI is the primary index of  $\Delta\phi$ -60. It measures whether a utilitarian calculation can be treated as a relatively stable welfare diagnosis, or whether it produces or intensifies subject-splitting by redistributing TCC in ways that create operational majorities and minorities.

```
USSI( $\pi$ ) = f(
  TCC_Concentration,
  Externalization_Asymmetry,
  ReEntry_Inequality,
  Closure_Risk,
  Benefit_Cost_Decoupling,
  CAPEI_Gap,
  Numerical_Operational_Subject_Gap
)
```

| Axis                              | Diagnostic question                                                            |
|-----------------------------------|--------------------------------------------------------------------------------|
| TCC Concentration                 | Is transition-completion cost concentrated on a specific coordinate?           |
| Externalization Asymmetry         | Does one coordinate reduce its cost by moving cost onto another?               |
| Re-entry Inequality               | Can cost-bearing coordinates contest and correct the attribution?              |
| Closure Risk                      | Can the cost-bearing coordinate survive without irreversible closure?          |
| Benefit-Cost Decoupling           | Are benefits and costs assigned to different coordinates?                      |
| CAPEI Gap                         | Are cost-editing authority and cost-bearing burden separated?                  |
| Numerical-Operational Subject Gap | Do numerical majority/minority labels fail to match operational editing power? |

| USSI band | Interpretation                                                                                                                         |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------|
| Low       | Utilitarian calculation does not strongly produce subject-splitting; cost distribution and re-entry access remain relatively balanced. |
| Medium    | The total may improve, but TCC concentration, benefit-cost decoupling, or re-entry inequality should be audited.                       |
| High      | The calculation likely produces or intensifies operational majority/minority boundaries by redistributing TCC.                         |
| Critical  | The calculation depends on cost-bearing coordinates that face high closure risk and weak re-entry access.                              |

## 9. Democratic Sovereignty and Operational Subjecthood

Democratic states often declare that the people are sovereign.  $\Delta\phi$ -60 does not reject this declaration at the level of legitimacy. However, it separates nominal sovereignty from operational subjecthood. The operational subject is not the one in whose name authority is declared, but the coordinate that can edit, redirect, authorize, or refuse the path of cost attribution.

*A right that cannot be exercised without unbearable TCC is formally present but operationally weak.*

| Democratic surface           | SΔφ-60 operational question                                                                       |
|------------------------------|---------------------------------------------------------------------------------------------------|
| Voting rights exist.         | Can citizens edit agenda-setting, policy design, and post-election cost attribution?              |
| Petitions are allowed.       | Does petitioning have a realistic path to institutional re-entry, or is it symbolic absorption?   |
| Courts are accessible.       | Is litigation TCC bearable for ordinary citizens and vulnerable groups?                           |
| Speech is legally protected. | What are the employment, stigma, surveillance, defamation, or retaliation costs of speaking?      |
| Civil society exists.        | Do organizations actually reduce cost-bearing groups' TCC and re-enter costs into editing paths?  |
| The people are sovereign.    | Who can actually redirect cost in law, budget, administration, platform rules, and public agenda? |

## 10. Civic Organizations: Re-entry or Theater

The existence of civic organizations does not prove that civil society is operational. SΔφ-60 distinguishes civic organization types by whether they reduce TCC for cost-bearing groups and create genuine institutional re-entry, or whether they merely absorb critique, stage diagnosis, or mobilize citizens without giving them editing authority.

| Type                                 | Operational description                                                                                                       | SΔφ diagnosis                                                 |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Genuine Re-entry Organization        | Records cost, lowers victim or citizen TCC, connects testimony to legal, policy, budgetary, or institutional editing.         | Operational civil society.                                    |
| Nominal / State-aligned Organization | Uses civic language while preserving existing authority and delaying cost re-entry.                                           | Cost-delay interface.                                         |
| Diagnostic Theater Organization      | Produces reports, statements, debates, or hearings without measurable editing or reduced recurrence cost.                     | Diagnosis without re-entry.                                   |
| Capture-Conversion Organization      | Begins as re-entry organization but becomes aligned with funders, parties, platforms, state agencies, or prestige incentives. | Re-entry path converted into managed channel.                 |
| Mobilization-only Organization       | Mobilizes citizens' time, risk, emotion, or labor while decision authority remains elsewhere.                                 | Citizens become cost-bearing resources, not editing subjects. |

*The existence of civic organizations does not prove that civil society is operational. A civic organization becomes operationally civic only when it lowers the TCC of cost-bearing groups and re-enters their cost into institutional editing paths.*

## 11. COSI — Civic Organization Subjecthood Index

COSI is an optional submodule for democratic and civil-society applications of SΔφ-60. It evaluates whether a civic organization opens subjecthood for cost-bearing groups or functions as a theater, buffer, or capture channel.

| Axis               | Question                                                                                      |
|--------------------|-----------------------------------------------------------------------------------------------|
| Cost Trace Capture | Does the organization record actual cost and harm rather than merely symbolic grievance?      |
| Re-entry Capacity  | Does recorded cost return to law, policy, budget, platform rules, or institutional procedure? |



|                                |                                                                                  |
|--------------------------------|----------------------------------------------------------------------------------|
| Edit Authority Access          | Can the organization access actual editing coordinates?                          |
| Victim / Citizen TCC Reduction | Does it reduce the cost of speaking, appealing, repairing, or organizing?        |
| Independence                   | Is it structurally independent from the authority whose cost path it audits?     |
| Feedback Integrity             | Can internal criticism and external evaluation change the organization itself?   |
| Outcome Trace                  | Does diagnosis leave a durable institutional or procedural trace?                |
| Theater Risk                   | Does activity replace re-entry with visibility, events, or symbolic performance? |

## 12. Reporting Template

A USSI report should avoid declaring a group guilty by label. It should specify the policy or decision, the affected coordinates, numerical scale, editing authority, benefit distribution, TCC distribution, externalization direction, closure risk, and re-entry access.

```

USSI Report
Policy/action  $\pi$ :
Declared objective:
Numerical majority/minority labels:
Operational editing authority:
Cost-bearing coordinates:
Benefit distribution:
TCC distribution:
Externalization matrix  $E_{i,j}$ :
CAPEI gap:
Re-entry inequality:
Closure risk:
UMR:
Observer position:
Exclusion register:
Measurement-induced cost:
Revision path:
Interpretation band: Low / Medium / High / Critical

```

### Example Report Pattern

```

Policy  $\pi$ : AI platform moderation and appeal policy
Declared objective: user safety and platform trust
Numerical majority: general users
Operational editing authority: platform operator and policy team
Cost-bearing coordinates: ordinary users, appeal applicants, outsourced moderators, AI output surface
Benefit distribution: platform trust and reduced visible risk
TCC distribution: appeal friction, verification burden, emotional labor, and correction cost shifted outward
Externalization direction: platform -> users/moderators/AI surface
CAPEI gap: high
Re-entry inequality: high
Closure risk: medium
USSI: high
Interpretation: The policy may serve safety goals, but it also produces operational subject-splitting by separating benefit control from cost-bearing coordinates.

```

## 13. Limits, Remainder, and Observer Register

SΔφ-60 inherits the measurement discipline of SΔφ-56 v1.1. Because subjecthood itself is not directly measurable, every USSI, CAPEI, or COSI report must preserve what remains unmeasured and must make the observer position visible.

| Required field             | Purpose                                                                                                                                      |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| UMR — Unmeasured Remainder | Preserve subjecthood, lived texture, political meaning, identity, suffering, trust, fear, and community residues not captured by the metric. |
| Observer Position          | State who measured: citizen, researcher, state agency, platform, civic group, court, journalist, affected group, or AI system.               |
| Exclusion Register         | List missing voices, missing costs, unavailable data, absent time horizons, and excluded nonhuman or future coordinates.                     |
| Measurement-Induced Cost   | Record whether the measurement itself creates stigma, retaliation, legal risk, defensive behavior, or new cost displacement.                 |
| Revision Path              | Specify how the report can be contested, corrected, updated, or withdrawn.                                                                   |

*A subject-splitting score without revision path becomes another instrument of subject-splitting.*

## 14. Closing Declaration

SΔφ-60 does not claim to measure the subject itself. It measures traces of cost-attribution path editing. Utilitarian calculation may appear to evaluate the happiness of an already-existing majority, but in some cases it first produces the boundary between majority and minority by redistributing TCC. The question is therefore not only whether the total is good, but who became the subject capable of editing cost, and who became the coordinate forced to bear it.

*The subject is not the name in whose favor the system speaks. The subject is the coordinate that can edit where cost returns.*

## Appendix A. Minimal Formula Set

TCC<sub>i</sub>(π) = transition-completion cost borne by coordinate i  
Benefit<sub>i</sub>(π) = benefit received by coordinate i  
E<sub>i,j</sub>(π) = TCC externalized from i to j  
CAPEI<sub>i</sub>(π) = cost-attribution path editing capacity of i  
USSI(π) = f(TCC concentration, externalization asymmetry, re-entry inequality, closure risk, benefit-cost decoupling, CAPEI gap, numerical-operational subject gap)

## Appendix B. Related SΔφ Documents

- SΔφ-01 — Minimal Core: Phase-Change Formalism of Existence
- SΔφ-06 — Responsibility as Irreversible Cost Internalization
- SΔφ-52 — Body as Non-Deferrable Cost Return
- SΔφ-56 — Transition Completion Cost, v1.1 reinforced
- SΔφ-58 — Cost Attribution Symmetry Index
- SΔφ-59 — Affective Cost Orientation Index